



# MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PECAIDS701A/PECAIML701A/PECICB701B Social Network Analysis

UPID : 007799

CS/B.TECH(N)/ODD/SEM-7/7799/2025-2026/1124

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin Indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

## Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[ 1 x 10 = 10 ]

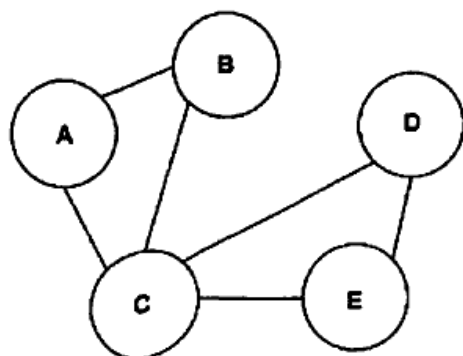
- (I) Is adjacency matrix of directed network symmetric?
- (II) What is the difference between undirected network and directed network?
- (III) What kind of matrix is the adjacency matrix for undirected network?
- (IV) What is centrality in social network analysis?
- (V) Is Email communication network a directed network?
- (VI) What are basic components of networks?
- (VII) What is density of a network?
- (VIII) What is clustering in social network analysis?
- (IX) What is eccentricity of a node in a network?
- (X) What is PageRank algorithm used for?
- (XI) Which centrality determines a node spreading information faster?
- (XII) What does network embeddedness depends on?

## Group-B (Short Answer Type Question)

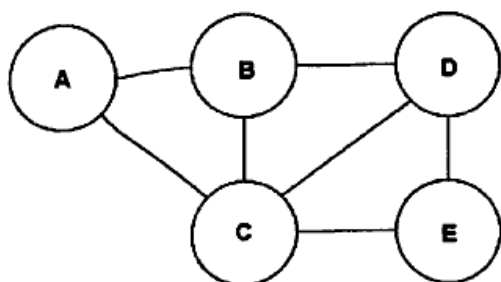
Answer any three of the following :

[ 5 x 3 = 15 ]

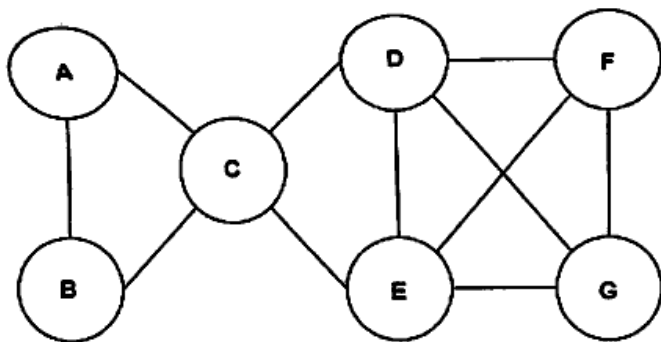
2. What are the three basic stages of spectral clustering? Explain them. [5]
3. Explain the EI index for homophily. [5]
4. Calculate the clustering coefficient of nodes A and C in the given network. [5]



5. Calculate the closeness centrality of all the nodes in the given network. [5]



6. Estimate the communities using Clique Percolation Method (CPM) in the given network (use clique size 3).

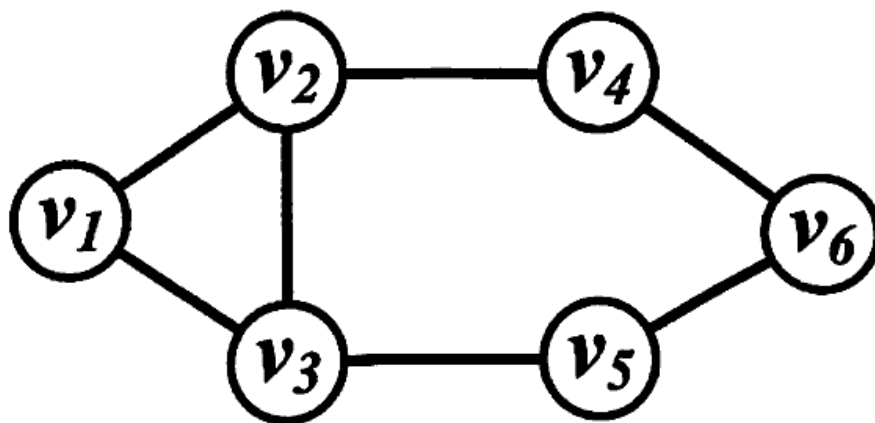


**Group-C (Long Answer Type Question)**

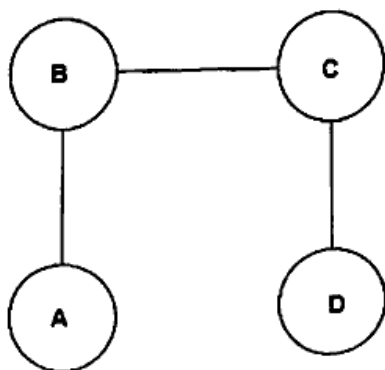
Answer any three of the following :

[ 15 x 3 = 45 ]

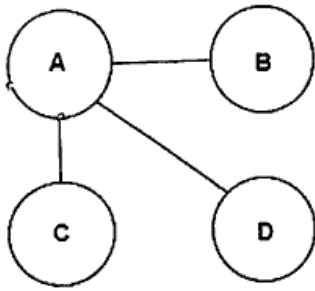
7. What are the types of relations or edges present in social networks. Explain with examples. [ 15 ]
8. What are the distance-based measures for describing the whole network? Define and explain every measure. [ 15 ]
9. (a) Write the formulas for Jaccard similarity and Cosine similarity. [ 2 ]  
(b) Calculate the Jaccard similarity between  $(v_4, v_3)$  and Cosine similarity between  $(v_2, v_5)$  for the given network. [ 4 ]



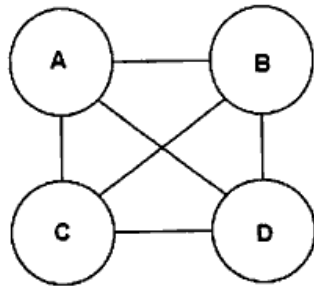
- (c) Calculate the modularity of the given network with the given communities. [ 9 ]



10. Explain the group-centric community detection methods. [ 15 ]
11. (a) Using closeness centrality find the more centralized network among the given networks. [ 8 ]



**Network 1**



**Network 2**

(b) Using degree centrality find the more centralized network among the given networks.

[ 7 ]

\*\*\* END OF PAPER \*\*\*